Abhirath **Anand**

abhirath@umich.edu theabhirath.com theabhirath theabhirath Abhirath Anand

EDUCATION

University of Michigan
Ann Arbor, USA

PhD in Bioinformatics — Jan. 2026 – Present

Working in the **Snitkin Lab** on genomic epidemiology to inform public health responses to infectious disease outbreaks, specifically in the context of healthcare-associated infections.

MS in Bioinformatics, **GPA**: 4.0/4.0 — Aug. 2024 – May 2026

Birla Institute of Technology and Science, Pilani (BITS Pilani)
Nov. 2020 – Jun. 2024

BE in Computer Science, **GPA**: 7.85/10 — Pilani, Rajasthan, India

Thesis: Worked with **Dr. Vinti Agarwal** in collaboration with **Dr. Ben Woodcroft**, Center for Microbiome Research (CMR), Queensland University of Technology on prediction models for salinity in Earth's microbial communities.

EXPERIENCE

Research Experience

Summer Intern
Jun. 2025 – Aug. 2025

University of Tennessee Health Science Center (UTHSC) — Memphis, Tennessee, USA

- Selected as part of the Biomedical Data Science Internship (BDSI) program in the Division of Biostatistics under the Department of Preventive Medicine.
- Worked with **Dr. Saunak Sen** and **Dr. Gregory Farage** on developing a Julia package designed to streamline working with metabolomics data.
- Developed separate packages for preprocessing, visualization and statistical analysis, further integrating them under a single umbrella package to serve as one-stop solution for metabolomics matrix data.

Summer Research Intern
May 2023 – Jul. 2023

Tata Institute of Fundamental Research (TIFR) — Mumbai, Maharashtra, India

- Selected as part of the Visiting Students' Research Program (VSRP) in the Department of Biological Sciences (DBS) at TIFR.
- Worked with **Dr. Amitesh Anand** on computational insights into the bioenergetics of microbes.
- Analysed Complex I of the electron transport chain in microbes, with a focus on its evolution and its spread across the microbial tree of life.

Undergraduate Research Assistant
Oct. 2022 – Dec. 2023

Microbial Interactions Lab, University of Southampton — Remote

- Worked with **Dr. Franklin Nobrega** on computational insights related to genomic plasticity in *Salmonella*.
- Focussed on using computational tools to analyse how genomic plasticity could impact factors like pathogenicity, antibiotic resistance, metal resistance and antiphage defence systems.

Undergraduate Research Projects
2022 – 2023

BITS Pilani — Pilani, Rajasthan, India

- Worked with **Dr. Balakumaran Chandrasekar** on analysing proteomics data using computational approaches, with a focus on phylogenetic orthology inference and protein-protein interaction modelling during the time period Aug. 2023 – Dec. 2023.
- Worked with **Dr. Vinti Agarwal** on graph machine learning algorithms for text classification during the time period Nov. 2022 – May 2023.
- Worked with **Dr. Tejasvi Alladi** in exploring machine learning applications in network security and in detecting unmanned aerial vehicles during the time period Jan. 2022 – Dec. 2022.

Engineering Experience

Google Summer of Code Contributor
May 2022 – Sep. 2022

Metalhead.jl, FluxML — Remote

- Worked on improving the deep learning for computer vision ecosystem in the Julia programming language, specifically in the FluxML ecosystem for deep learning with a focus on the Metalhead.jl library.
- Refactored the Metalhead.jl library to incorporate more modern training techniques and newer-age models.
- Developed a general purpose API for models to make them hackable and allow machine learning researchers and engineers to start experimentation easily.

Summer Intern
May 2022 – Jul. 2022

GERMI (Gujarat Energy Research and Management Institute) — Remote

- Worked on extreme learning machines (ELMs) and their uses.
- Explored using ELMs in tandem with autoencoders for online condition monitoring systems in industrial environments.

PUBLICATIONS

Hidden Markov Model-Based Prokaryotic Genome Space Mining Reveals the Widespread Pervasiveness of Complex I and Its Potential Evolutionary Scheme
Akshay Shirsath, Snehal V Khairnar, **Abhirath Anand**, Divya M Prabhakaran, and Amitesh Anand
Genome Biology and Evolution, 2025 · doi:10.1093/gbe/evaf154

Comprehensive blueprint of *Salmonella* genomic plasticity identifies hotspots for pathogenicity genes
Simran Krishnakant Kushwaha, Yi Wu, Hugo Leonardo Avila, **Abhirath Anand**, Thomas Sicheritz-Pontén, Andrew Millard, Sandhya Amol Marathe, and Franklin L. Nobrega
PLOS Biology, 2024 · doi:10.1371/journal.pbio.3002746

TALKS

BigRiverMetabolomics.jl: A Julia Framework for Scalable Metabolomics Data Analysis — Jul. 2025
Webinar, Division of Biostatistics, UTHSC — Remote
Presented the architecture and design principles of an open-source, end-to-end Julia package for metabolomics data analysis, covering preprocessing, statistical analysis and visualisation, and demonstrated it on a real-world metabolomics dataset.

Introduction to Machine Learning and Deep Learning — Mar. 2023
Datageddon, APOGEE — Pilani, Rajasthan, India
Invited by the Computer Science Association (CSA) at BITS Pilani to give a two-part workshop at the Datageddon hackathon during APOGEE, the institute's techfest. Introduced first-year students to the basics of machine learning and deep learning, with a hands-on walkthrough of training a multilayer perceptron to classify handwritten digits.

POSTERS

hospitraceR: An R Package for WGS-Based Genomic Epidemiological Analysis of Healthcare-Associated Infections — Oct. 2026
ASM BIG 2026 — Washington, D.C., USA
Presented hospitraceR, which combines whole-genome sequencing data with patient location metadata to infer, characterise and compare transmission clusters of healthcare-associated infections, applied to 889 KPC-*Klebsiella pneumoniae* isolates from four Chicago long-term acute care hospitals.

BigRiverMetabolomics.jl: A Julia Framework for Scalable Metabolomics Data Analysis — Jul. 2025
Community Science Exhibition, Rhodes College — Memphis, Tennessee, USA
Showcased the package's data preparation, visualisation and statistical analysis features to a broad, non-academic audience, using the "Even Better Poster" format.

WORKSHOPS ATTENDED

Simons Centre for Living Systems, NCBS — 19 – 28 Jun. 2023
"Physics of Life" Monsoon School — Bangalore, India
- Attended a 10 day "Monsoon School" organised by the **Simons Centre for Living Systems at the National Centre for Biological Sciences (NCBS)** designed to introduce students from quantitative backgrounds to interdisciplinary research in biology.
- Chosen as part of a **select cohort of 50 students** from across India.
- Attended lectures and hands-on sessions on a variety of topics in biology.
- Worked on a project to model predator-prey interactions in real time using the Viscek model.

IIIT Hyderabad — 18 July 2022 – 19 Aug 2022
Summer School on Artificial Intelligence — Remote
- Attended a month long summer school on Artificial Intelligence organised by the Centre for Visual Information Technology at the International Institute of Information Technology, Hyderabad **(CVIT, IIIT Hyderabad)**.
- Attended lectures and hands-on sessions on a variety of topics in AI, including machine learning, deep learning, and reinforcement learning.
- Was exposed to a variety of application areas of AI, including computer vision, natural language processing, and robotics.

PROJECTS

2023 **Compiler for toy language "ERPLAG"**, GitHub repository — Compiler Design
2022 **FluxML/Metalhead.jl**, GitHub repository — Computer Vision

CLUBS AND SOCIETIES

Sally Robotics
May 2021 – Oct. 2023
Team Lead and Computer Vision Researcher
Pilani, Rajasthan, India

- Joined Sally Robotics as a Computer Vision Researcher working on problems related to self-driving cars and autonomous navigation in the context of Indian roads.
- Was promoted to **Team Lead** in May 2022 for the duration of the academic year 2022-23. Headed a team of 20+ members across four subsystems (Computer Vision, SLAM, Navigation and Mechanical) working on problems related to autonomous driving.
- Mentored and guided students in the team, specifically in computer vision and deep learning, and facilitated regular discussions on research papers and new developments in the field.
- Was responsible for procuring funding for the team worth ₹150k from university grants.

BITS Model United Nations Society (BITSMUN)
Nov. 2020 – Nov. 2021
Member
Pilani, Rajasthan, India

- As part of the Model United Nations Society, engaged in weekly discussions on international affairs.
- Participated in several Model United Nations conferences conducted by various colleges across India.

HONORS & AWARDS

2020 **All India Rank (AIR) 1183**, Joint Entrance Examination (JEE) Mains — India
2020 **All India Rank (AIR) 1640**, Joint Entrance Examination (JEE) Advanced — India
2019 **All India Rank (AIR) 389**, Kishore Vaigyanik Protshahan Yojana (KVPY) SA — India
2018 **NTS Scholarship**, National Talent Search Scholarship — India

SKILLS

Programming C, Python, Julia, Java, LaTeX
Technical Skills Machine Learning, Deep Learning, Computer Vision, Bioinformatics
Languages English, Hindi, Tamil